

Linux Hardening Audit Tool

Cybersecurity Internship Project Report

Platform	Language	Final Score	Status
Kali Linux VM	Bash	60/100	Needs Improvement

Abstract

This project developed a lightweight Bash-based Linux security auditing tool that automates common hardening checks on a Kali Linux system. The script inspects firewall policies, user privileges, important file permissions, SSH settings, package updates, password aging, running services, and basic rootkit indicators. It classifies results as PASS, WARNING, or INFO, calculates a custom security score, and produces hardening recommendations. The goal was to create a simple defensive-security tool that is easy to understand, run, and explain in a SOC or Linux security context.

Introduction

Linux hardening reduces unnecessary exposure by reviewing security-sensitive configuration and identifying weak settings before they are abused. Instead of checking each item manually, this project combines several Linux security checks into one script. The tool is educational and lightweight; its score is a custom project metric and is not an official CIS Benchmark compliance score.

Tools Used

- Kali Linux running in Oracle VirtualBox
- Bash shell scripting
- iptables, systemctl, ss, awk, grep, stat, find
- Linux configuration files such as /etc/passwd, /etc/shadow, /etc/ssh/sshd_config, and /etc/login.defs

Steps Involved in Building the Project

- **Version 1:** Collected raw system information, current-user details, listening ports, running services, firewall rules, file permissions, UID 0 users, and SSH configuration.
- **Version 2:** Added automated PASS/WARNING logic, validation of firewall and file-permission settings, SSH checks, and a custom security score.
- **Version 3:** Added package-update checks, password-aging analysis, detection of risky legacy services, SUID binary inspection, and basic rootkit indicators.
- **Final version:** Added hardening recommendations and saved the audit output to a text report for documentation and GitHub submission.

```
[5] FIREWALL CHECK
[WARNING] INPUT firewall policy is: ACCEPT
[WARNING] FORWARD firewall policy is: ACCEPT

[6] IMPORTANT FILE PERMISSIONS
[PASS] /etc/passwd permissions: 644
[PASS] /etc/shadow permissions: 640

[7] USERS WITH UID 0
[PASS] Only root has UID 0

[8] SSH CONFIGURATION
[INFO] PermitRootLogin is not explicitly configured
[INFO] PasswordAuthentication is not explicitly configured

[9] SYSTEM UPDATE CHECK
[WARNING] 836 package update(s) available

[10] PASSWORD POLICY CHECK
[INFO] Password maximum age: 99999 days
[INFO] Password minimum age: 0 days
[WARNING] Password maximum age is greater than 90 days

[11] RISKY SERVICES CHECK
[PASS] No common risky legacy services detected

[12] BASIC ROOTKIT INDICATOR CHECK
[INFO] SUID binaries detected: 28
[INFO] chkrootkit is not installed
[INFO] Current check is limited to basic indicators
```

Figure 1. Expanded audit checks showing firewall warnings, update status, password policy, service checks, and rootkit indicators.

Key Findings and Audit Result

Check	Result
Firewall	INPUT and FORWARD default policies were ACCEPT, so the tool flagged them for review.
File permissions	/etc/passwd was 644 and /etc/shadow was 640; both checks passed.
Privilege check	Only root had UID 0; this check passed.
SSH	PermitRootLogin and PasswordAuthentication were not explicitly configured, so they were reported as informational findings.
Updates	836 package updates were detected at the time of testing.
Password policy	PASS_MAX_DAYS was 99999 days, which triggered a warning because the project threshold was 90 days.
Legacy services	No common risky legacy services such as telnet, ftp, rsh, or rlogin were detected.
Rootkit indicators	28 SUID binaries were counted and chkrootkit was not installed; the script clearly labels this as only a basic indicator check.

```
[13] HARDENING RECOMMENDATIONS
[RECOMMEND] Review INPUT firewall rules and consider a restrictive default policy.
[RECOMMEND] Review FORWARD firewall rules and disable forwarding if not required.
[RECOMMEND] Install pending security and package updates.
[RECOMMEND] Reduce PASS_MAX_DAYS in /etc/login.defs to 90 days or less.
[RECOMMEND] Periodically review SUID binaries for unexpected changes.
[RECOMMEND] Consider using a dedicated rootkit detection tool for deeper scans.

[14] SECURITY SCORE
Security Score: 60/100
Overall Status: NEEDS IMPROVEMENT

AUDIT COMPLETE
```

Figure 2. Final hardening recommendations and custom security score of 60/100.

Conclusion

The project successfully automated multiple Linux hardening checks in a single Bash script and converted raw system information into actionable security findings. The final tool identified permissive firewall policies, a large number of pending updates, and an overly long password-aging setting while also confirming secure file permissions, a single UID 0 account, and the absence of common risky legacy services. The project demonstrates practical defensive-security concepts including configuration auditing, privilege review, service monitoring, scoring, and remediation guidance. Future improvements could include fuller CIS Benchmark mapping, deeper SSH-policy evaluation, dedicated vulnerability checks, and more advanced rootkit detection.

Note: The security score is based on custom project rules for educational use and should not be treated as an official compliance rating.